An Ensemble Hybrid Approach for Pre-Symptomatic Plant Disease Prediction Using Biosensor and Soil Data

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***Abstract*—Plant diseases cause 20–40% of global crop losses annually, posing a severe and growing threat to food security, agricultural sustainability, and farmer livelihoods worldwide. Conventional detection methods depend entirely on visual inspection by trained agronomists, which identifies disease only after visible symptoms have already manifested on the plant – at which point significant tissue damage has occurred and large-scale prevention is no longer feasible. This proposes a pre-symptomatic early warning system that leverages continuous biosensor and soil sensor readings to predict plant health status – classified as Healthy, Moderate Stress, or High Stress – before any visible symptoms appear. A hybrid stacking ensemble model combining Random Forest (RF) and Support Vector Machine (SVM) as base learners, with Logistic Regression as the meta-learner, is applied to a dataset of 1,200 sensor readings collected from 10 plants at 6-hour intervals over approximately one month. The dataset encompasses 11 environmental and soil features: Soil Moisture, Ambient Temperature, Soil Temperature, Humidity, Light Intensity, Soil pH, Nitrogen Level, Phosphorus Level, Potassium Level, Chlorophyll Content, and Electrochemical Signal. The proposed hybrid stacking model achieved approximately 98% prediction confidence on held-out samples, outperforming the individual Random Forest classifier (approximately 98%) and SVM classifier (approximately 93%). A continuous Disease Risk Score in the range $[0, 100]$ was additionally formulated to quantify disease severity beyond categorical class labels, enabling more actionable farmer guidance. Experimental results confirm that Chlorophyll Content and Electrochemical Signal are the strongest predictors of plant stress, consistent with known plant physiology. The system requires no imaging equipment and operates fully automatically via IoT biosensors, making it practically deployable in real-world smart farming environments.**

***Index Terms*—plant disease prediction, random forest, support vector machine, hybrid ensemble, stacking classifier, biosensor data, disease risk score, precision agriculture, IoT, early detection, electrochemical signal**

I. Introduction

Plant diseases are among the most critical and persistent threats to global agricultural productivity and food security. According to the Food and Agriculture Organization (FAO) of the United Nations, pathogens, pests, and plant diseases collectively destroy an estimated 20 to 40 percent of global crop production every year [1]. The resulting economic losses amount to hundreds of billions of dollars annually, and the human consequences are equally severe, threatening the livelihoods of approximately 500 million smallholder farming families who depend on agriculture for their income and food supply. In rapidly developing agricultural economies such as India, where farming contributes significantly to the national gross domestic product and employs a large segment of the rural population, the impact of undetected plant diseases is disproportionately severe.

The conventional method for plant disease detection involves physical observation by trained agronomists or experienced farmers who visually inspect crops for characteristic symptoms such as leaf discoloration, chlorotic spots, wilting, necrotic lesions, or abnormal growth patterns. This approach suffers from a fundamental and unavoidable limitation: the disease is only observable after it has already caused measurable physiological damage to the plant. By the time symptoms are visible, the pathogen has typically established itself within plant tissue, triggered systemic stress responses, and in many cases, dispersed spores or bacterial cells to neighboring plants. At this advanced stage, chemical treatment can slow further progression but cannot reverse the damage already done. Furthermore, visual inspection is inherently labor-intensive, subjective, limited in scale, and entirely dependent on the availability of expert knowledge – a resource that is scarce in many rural agricultural communities.

The recent proliferation of Internet of Things (IoT) technology, low-cost sensing hardware, and wireless communication infrastructure has created a fundamentally different opportunity for plant health monitoring. Modern biosensors can continuously and automatically measure a wide range of environmental and soil parameters – including temperature, relative humidity, soil moisture, pH, macro-nutrient concentrations, light intensity, chlorophyll content via reflectance, and electrochemical stress signals emitted by plant root zones – at high temporal frequency and without any human intervention. These parameters are not merely indirect correlates of disease; they reflect the physiological and biochemical state of the plant and its immediate environment with established scientific relationships to disease susceptibility and stress onset.

Machine learning algorithms have demonstrated strong per-

formance across diverse agricultural classification tasks. Ensemble methods such as Random Forest [5] and kernel-based classifiers such as Support Vector Machine [6] are particularly well-suited to structured tabular sensor data. When trained on historical biosensor readings labeled with corresponding plant health outcomes, such models can learn to recognize early stress signatures that precede visible disease symptoms by days or even weeks, enabling truly pre-symptomatic intervention.

Existing work in plant disease prediction has been dominated by image-based approaches that apply convolutional neural networks (CNN) to photographs of plant leaves [2] [3]. While these methods achieve impressive classification accuracy on benchmark datasets, they are fundamentally reactive – requiring visible symptoms to already be present – and impose significant practical barriers to field deployment, including the need for high-resolution cameras, consistent lighting conditions, large amounts of labeled image data, and substantial computational resources for inference.

The present work proposes an alternative paradigm: *sensor-driven, pre-symptomatic plant health prediction* using a hybrid ensemble model that combines the complementary strengths of Random Forest and SVM through stacked generalization [7]. The main contributions of this paper are as follows:

- A hybrid RF-SVM stacking ensemble classifier for plant health prediction achieving approximately 98% prediction confidence on held-out biosensor data.
- A continuous Disease Risk Score $R(x) \in [0, 100]$ derived from model probability outputs to quantify disease severity beyond categorical labels, enabling tiered farmer alert systems.
- The incorporation of Electrochemical Signal and Chlorophyll Content as predictive features, which are rarely employed in existing plant disease machine learning literature despite their strong physiological relevance.
- A rigorous three-way model comparison between Random Forest, SVM, and the proposed Hybrid Stacking model using accuracy, precision, recall, F1-score, and cross-validation.
- A fully automated, camera-free, IoT-deployable prediction pipeline that operates continuously without requiring human inspection or expert agronomist involvement.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and preprocessing pipeline. Section IV presents the proposed methodology in detail. Section V reports experimental results and analysis. Section VI concludes the paper and outlines future research directions.

II. Related Work

A. Image-Based Plant Disease Detection

The majority of published research on automated plant disease detection has focused on image-based deep learning approaches. Mohanty et al. [2] demonstrated the potential of deep convolutional neural networks by training on the PlantVillage dataset comprising 54,306 labeled leaf images across 26 diseases and 14 crop species, achieving 99.35% classification accuracy under controlled conditions. However, performance degraded significantly when tested on field images, highlighting the domain gap between laboratory-quality photographs and real-world agricultural conditions.

Ferentinos [3] evaluated multiple CNN architectures for plant disease recognition from leaf photography and confirmed high accuracy while noting that image-based systems are inherently constrained to detecting disease only after visible symptoms have appeared. Sladojevic et al. [10] applied deep neural networks to leaf image classification for multiple crop diseases, achieving strong results on a curated dataset but again requiring post-symptomatic images as input. None of these approaches address pre-symptomatic detection or operate without visual input.

B. Sensor-Based and Environmental Approaches

Pantazi et al. [8] proposed supervised machine learning for plant stress detection using soil and crop sensor data, demonstrating that Random Forest achieves strong classification performance on structured tabular data from agricultural sensors. Their work established the viability of non-image machine learning for plant health monitoring but did not employ hybrid ensemble architectures or electrochemical sensing.

Ahmed et al. [9] compared SVM and Random Forest classifiers on datasets containing environmental parameters including temperature, humidity, and soil nutrient concentrations, achieving over 90% accuracy in both cases. Their comparative analysis confirmed that Random Forest generally outperforms SVM in multi-class agricultural classification tasks, motivating the use of both as components in the hybrid ensemble proposed in the present work.

C. Hybrid and Ensemble Approaches

Ensemble methods have been widely applied to improve classification accuracy across many domains. Stacked generalization, introduced by Wolpert [7], trains a meta-learner on the out-of-fold predictions of base models, allowing the meta-learner to optimally combine their complementary strengths. This approach has been applied in medical diagnosis, financial forecasting, and fault detection systems, but its application to pre-symptomatic plant disease prediction from biosensor data remains largely unexplored in the existing literature.

The present work addresses this gap by proposing a stacking ensemble specifically designed for biosensor-driven plant health classification, incorporating the novel Disease Risk Score formulation and SHAP-based explainability to bridge the gap between model output and actionable farmer guidance.

III. Dataset and Preprocessing

A. Dataset Description

The dataset used in this study consists of 1,200 biosensor readings collected from 10 monitored plants over approximately one month, with sensor readings recorded at regular 6-hour intervals. All data were generated by IoT-connected

soil and environmental biosensors, producing a fully structured tabular dataset with no image components. The complete absence of missing values was confirmed via Pandas null checks prior to preprocessing.

Table I summarizes the 13 original dataset columns. Following preprocessing, Timestamp and Plant_ID were removed, yielding 11 numerical input features and one categorical target label.

TABLE I
DATASET FEATURES, UNITS, AND ROLES

Feature	Unit	Role
Timestamp	–	Dropped
Plant_ID	–	Dropped
Soil Moisture	%	Input
Ambient Temperature	°C	Input
Soil Temperature	°C	Input
Humidity	%	Input
Light Intensity	Lux	Input
Soil pH	–	Input
Nitrogen Level	mg/kg	Input
Phosphorus Level	mg/kg	Input
Potassium Level	mg/kg	Input
Chlorophyll Content	mg/m^2	Input
Electrochemical Signal	mV	Input
Plant_Health_Status	–	Target

The target variable *Plant_Health_Status* contains three classes representing disease risk levels. Class distribution is approximately balanced across the three categories, confirmed via exploratory data analysis.

B. Target Label Encoding

Labels were encoded using a manual severity-ordered mapping to preserve the ordinal relationships among health states: Healthy = 0, Moderate Stress = 1, High Stress = 2. This explicit encoding was preferred over automatic LabelEncoder to avoid alphabetically-ordered assignment (which would incorrectly place High Stress before Moderate Stress), ensuring that the label ordering reflects true disease severity progression.

C. Feature Scaling

The 11 input features operate on vastly different numerical scales. For example, Light Intensity ranges from 0 to over 1,000 Lux, while Soil pH ranges from 4.5 to 8.5, and Electrochemical Signal ranges from 0 to approximately 2 mV. Without normalization, distance-based algorithms such as SVM would be dominated by high-magnitude features. StandardScaler was applied to transform all features to zero mean and unit variance:

$$x' = \frac{x - \mu}{\sigma} \tag{1}$$

where μ and σ are the feature mean and standard deviation computed on the training set and applied to both training and test sets to prevent data leakage.

D. Train-Test Split

The dataset was partitioned into 80% training (960 samples) and 20% testing (240 samples) using stratified random sampling with `random_state=42` to ensure reproducibility. Stratification ensures that the class proportion of each health status category is preserved in both the training and testing subsets, preventing evaluation bias from class imbalance.

IV. PROPOSED METHODOLOGY

A. Individual Base Models

1) Random Forest Classifier: Random Forest [5] is a bagging ensemble that trains T decision trees, each on a bootstrap sample of the training data. At each split node, only a random subset of $\sqrt{p}$ features (where p is the total number of features) is considered, introducing decorrelation between trees. The final prediction is determined by majority vote across all trees:

$$\hat{y}_{RF}(x) = \arg\max_{c} \sum_{t=1}^{T} \mathbf{1}\left[h_t(x) = c\right] \tag{2}$$

where $h_t(x)$ is the prediction of the t-th tree. Random Forest also produces feature importance scores ϕ_i for each feature i, computed as the mean decrease in Gini impurity across all trees and all splits where feature i is used. Configuration: `n_estimators=100`, `random_state=42`, `n_jobs=-1`.

2) Support Vector Machine: SVM [6] finds the maximum-margin hyperplane separating classes in a kernel-induced feature space. The Radial Basis Function (RBF) kernel was selected via preliminary evaluation:

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right) \tag{3}$$

The RBF kernel implicitly maps inputs to an infinite-dimensional space, enabling non-linear decision boundaries in the original feature space. Multi-class classification was handled using the one-vs-one strategy. `probability=True` was set to enable soft probability outputs required by the stacking meta-learner. Configuration: `kernel=rbf`, `C=1.0`, `gamma=scale`.

B. Hybrid Stacking Ensemble

The proposed hybrid model employs stacked generalization [7], also referred to as stacking, to combine the complementary predictions of Random Forest and SVM. The stacking architecture operates in two levels as illustrated below.

Level 0 (Base Learners): Random Forest and SVM are trained on the training set using 5-fold cross-validation. For each fold, each base model generates out-of-fold probability predictions for the validation samples. This produces a new meta-feature matrix $\mathbf{M} \in \mathbb{R}^{n \times 2k}$ where n is the number of training samples and $k = 3$ is the number of classes. Each row of $\mathbf{M}$ contains the concatenated probability outputs of RF and SVM for that sample.

Level 1 (Meta-Learner): Logistic Regression is trained on the meta-feature matrix $\mathbf{M}$. The meta-learner learns the

optimal weighting of the base model outputs. Formally, the final prediction is:

$$\hat{y}(x) = g\left(f_{RF}(x),\ f_{SVM}(x)\right) \tag{4}$$

where $f_{RF}(x)$ and $f_{SVM}(x)$ are the probability vectors from Random Forest and SVM respectively, and g is the Logistic Regression meta-learner. Using out-of-fold predictions during meta-training prevents data leakage and avoids overfitting to base model errors on seen data.

C. Disease Risk Score

To provide actionable guidance beyond categorical predictions, a continuous Disease Risk Score $R(x) \in [0, 100]$ is derived from the hybrid model's class probability output $\mathbf{p}(x) = [p_0(x),\ p_1(x),\ p_2(x)]$ for classes Healthy, Moderate Stress, and High Stress respectively:

$$R(x) = 50 \cdot p_1(x) + 100 \cdot p_2(x) \tag{5}$$

This formulation assigns a risk weight of 50 to Moderate Stress probability and 100 to High Stress probability, yielding a score of 0 for a perfectly healthy plant and 100 for a plant with certainty of High Stress. The score is continuous and monotonically increasing with disease severity, enabling the following tiered farmer alert system: Safe ($R \leq 30$), Monitor ($30 < R \leq 60$), Warning ($60 < R \leq 80$), and Critical ($R > 80$).

D. Explainability via Feature Importance

To ensure the system is interpretable and trusted by end users, feature importance scores are extracted from the Random Forest base model. These scores quantify the contribution of each sensor reading to the prediction, allowing farmers and agronomists to understand which environmental or soil factors are driving the health status prediction. This interpretability layer is a critical component for real-world adoption of AI-based agricultural advisory systems.

E. Implementation Details

All experiments were implemented in Python 3.13 using the Scikit-learn library [4]. The complete pipeline was organized across 9 modular Jupyter notebooks: data loading, preprocessing, exploratory data analysis, Random Forest training, SVM training, hybrid stacking, model comparison, risk score and explainability, and prediction. All trained models and the StandardScaler were serialized using Joblib for deployment. The complete codebase is publicly available at: *github.com/SreeHarshitha-12/plant_disease_detection*.

V. Experimental Results and Discussion

A. Model Performance Comparison

Table II summarizes the classification performance of all three models evaluated on the 240-sample held-out test set. The hybrid stacking model outperforms both individual base classifiers across all reported metrics.

The improvement of the hybrid stacking model over the best individual model (Random Forest at ∼96%) represents a meaningful gain in a medical-style classification scenario where misclassification of a High Stress plant as Healthy could result in undetected disease progression and significant crop loss. The SVM classifier, while showing lower overall recall compared to Random Forest, demonstrated higher precision for the High Stress class, suggesting that it captures a different aspect of the decision boundary. The stacking meta-learner successfully learns to leverage both behaviors.

TABLE II
Model Performance Comparison on Test Set

Model	Acc.	Prec.	Recall	F1
Random Forest	∼96%	High	High	High
SVM (RBF Kernel)	∼93%	High	Moderate	Moderate
Hybrid (Stack)	**∼98%**	**Highest**	**Highest**	**Highest**

B. Class-wise Performance Analysis

Across all three classes, the hybrid model demonstrated the most balanced performance. The Healthy class was classified with the highest confidence in all models, consistent with the expectation that normal physiological ranges are more distinct from stressed states. The Moderate Stress class posed the greatest challenge, as it occupies the boundary region between Healthy and High Stress in feature space. The stacking ensemble's improvement over individual models was most pronounced for Moderate Stress classification, confirming that the complementary nature of RF and SVM is particularly beneficial at class boundaries.

C. Prediction Confidence on New Samples

Verification on representative held-out sensor samples confirmed the high operational confidence of the deployed hybrid model. A sample representing favorable growing conditions (Soil Moisture: 60%, Humidity: 45%, Chlorophyll Content: 38 mg/m^2, Electrochemical Signal: 0.3 mV) was correctly classified as *Healthy* with 99.24% model confidence. A sample representing severe stress conditions (Soil Moisture: 15%, Humidity: 88%, Chlorophyll Content: 12 mg/m^2, Electrochemical Signal: 0.95 mV) was correctly classified as *High Stress* with 98.85% confidence. These results demonstrate that the hybrid model produces highly reliable predictions across the full range of health states.

D. Feature Importance Analysis

Feature importance scores extracted from the Random Forest base model revealed a clear hierarchy of predictive value among the 11 sensor features. Chlorophyll Content emerged as the single most important predictor, followed closely by Electrochemical Signal. Soil Moisture and Humidity ranked third and fourth respectively, with Soil pH and Nitrogen Level contributing moderate importance. Light Intensity, Ambient Temperature, Soil Temperature, Phosphorus Level, and Potassium Level contributed comparatively less predictive power in isolation, though they contribute collectively to the model's decision boundaries.

These results are physiologically consistent with known plant stress biology. Chlorophyll degradation is one of the earliest molecular-level responses to plant stress, occurring well before any visible color change is detectable by the human eye. The Electrochemical Signal reflects ionic imbalances and stress-induced electrical activity in plant root zones, which are known to precede visible disease manifestation by 24–72 hours. This confirms that the proposed sensor suite is well-designed for pre-symptomatic detection.

E. Cross-Validation Analysis

Five-fold cross-validation was performed on the training set to assess model generalization beyond a single train-test split. Table III summarizes the mean and standard deviation of cross-validation accuracy for each model.

TABLE III
5-FOLD CROSS-VALIDATION RESULTS

Model	Mean CV Acc.	Std. Dev.
Random Forest	∼95.8%	±1.2%
SVM (RBF)	∼92.5%	±1.5%
Hybrid (Stacking)	∼97.5%	±0.8%

The hybrid stacking model not only achieves the highest mean CV accuracy but also the lowest standard deviation, indicating that it is more stable and less sensitive to the particular composition of the training fold. This is a key advantage of ensemble methods: by combining multiple models, variance is reduced without a corresponding increase in bias.

F. Disease Risk Score Validation

The Disease Risk Score was validated on a sample of 50 test instances across all three health classes. Healthy plants consistently received scores below 20, Moderate Stress plants received scores in the range 35–65, and High Stress plants received scores above 75. This monotonic alignment between categorical health status and continuous risk score confirms that the formulation in Equation 5 is well-calibrated and provides meaningful quantitative guidance beyond the categorical prediction alone.

G. Comparison with Related Work

Table IV positions the proposed approach relative to key related works in plant disease prediction.

TABLE IV
COMPARISON WITH RELATED APPROACHES

Work	Input	Pre-Symp.	Acc.
Mohanty et al. [2]	Images	No	99.35%
Ferentinos [3]	Images	No	∼97%
Pantazi et al. [8]	Sensors	No	∼90%
Ahmed et al. [9]	Env.	No	>90%
Proposed	**Sensors**	**Yes**	**∼98%**

The proposed system is the only approach in this comparison that achieves both pre-symptomatic prediction capability and competitive classification accuracy without relying on image data, demonstrating the practical advantage of the sensor-driven hybrid ensemble approach.

VI. CONCLUSION

This paper presented a hybrid ensemble machine learning system for pre-symptomatic plant disease prediction using biosensor and soil sensor data. The proposed stacking classifier combining Random Forest and SVM as base learners with Logistic Regression as a meta-learner achieved approximately 98% prediction confidence on held-out sensor samples, outperforming individual Random Forest (∼96%) and SVM (∼93%) classifiers. Cross-validation confirmed robust generalization with a mean accuracy of approximately 97.5% and the lowest variance among all evaluated models.

Five novel contributions distinguish this work from the existing literature. First, pre-symptomatic disease prediction is achieved purely from sensor data without any visual inspection or imaging requirement. Second, the hybrid RF-SVM stacking ensemble exploits the complementary decision boundaries of both base learners, achieving accuracy superior to either individual model. Third, a continuous Disease Risk Score formulation provides tiered, actionable severity guidance beyond categorical class labels. Fourth, Electrochemical Signal and Chlorophyll Content are identified and validated as powerful predictive features with strong physiological grounding. Fifth, the complete pipeline is modular, reproducible, version-controlled, and publicly available on GitHub, facilitating replication and extension by the research community.

Feature importance analysis confirmed that Chlorophyll Content and Electrochemical Signal are the strongest predictors of plant stress, consistent with their known roles as early biomarkers of cellular stress in plant physiology. The system eliminates dependency on cameras, expert visual inspection, and favorable imaging conditions, making it practically deployable in real-world smart farming environments through integration with IoT sensor networks and Flask/FastAPI web services.

Future research will pursue several directions. Expanding the dataset to include multiple crop species, diverse geographic regions, and a wider range of disease types will improve model generalization. Applying LSTM or Transformer-based architectures to model the temporal evolution of sensor readings may capture stress onset dynamics that a static feature-based classifier cannot represent. Developing a farmer-facing mobile application with real-time alert integration will bridge the gap between the ML model and agricultural end users. Federated learning approaches will be explored to enable privacy-preserving collaborative model training across multiple farms. Finally, extending the output from three-class health status prediction to specific disease identification – fungal, bacterial, or viral – represents a significant and impactful research direction.

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